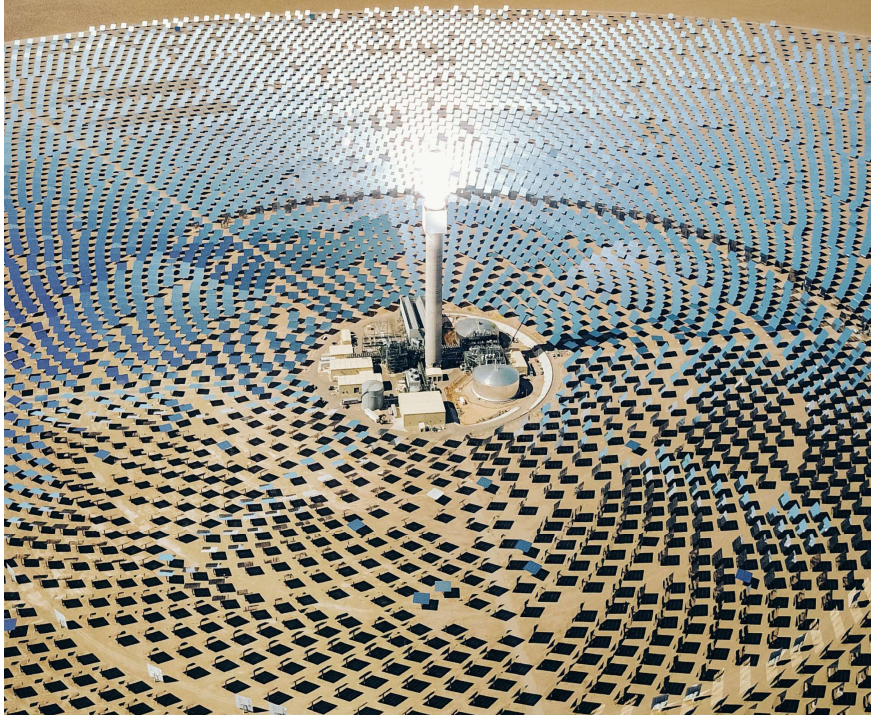


8TH
EDITION

K.A. STROUD ENGINEERING MATHEMATICS

WITH DEXTER J. BOOTH



PROGRAMME 5

MATRICES: SYSTEMS OF LINEAR EQUATIONS

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Worked examples and exercises are in the text

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Programme 5: **Matrices**

Solution of a set of linear equations

- 1. Inverse matrix method**
- 2. Gaussian elimination method**
- 3. Reduced row echelon form**
- 4. Cramer's rule**
- 5. General solution of a set linear equations**

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Worked examples and exercises are in the text

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Programme 5: Matrices

Learning outcomes

- ✓ Solve a set of linear equations using inverse matrices
- ✓ Use the Gaussian elimination method to solve a set of linear equations
- ✓ Use Gaussian elimination to bring a matrix to row echelon form (or reduced row echelon form) and hence solve it
- ✓ Use Gaussian elimination to find the inverse of a matrix by bringing it to reduced row echelon form i.e the identity matrix
- ✓ Use Cramer's rule to solve a set of linear equations
- ✓ Write the general solution of a set of linear equations (where a solution may or may not exist)

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Worked examples and exercises are in the text



Motivation:

Engineers need to solve **sets of linear equations** when analysing circuits, performing structural analysis and when analysing big data.



Figure 1: Truss structure support of roof in Hamburg Airport.

In general, structural analysis of trusses of any type can be carried out using a matrix method. The use of a matrix based approach makes it possible to treat systems with many variables in a general structured way.

Programme 5: Matrices

How can we solve a whole series of interlinked equations?

$$\begin{array}{ccccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \dots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \dots & + & a_{2n}x_n & = & b_2 \\ \vdots & & & & \vdots & & & & \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \dots & + & a_{mn}x_n & = & b_m \end{array}$$

- given all of the real-number values a_{ij} and b_i
- find values $x_1, \dots, x_n$ that satisfy **all** the equations at once

Solution of a set of linear equations

The set of n simultaneous linear equations in n unknowns

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n &= b_1 \\a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n &= b_2 \\ \vdots & \\a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \dots + a_{nn}x_n &= b_n\end{aligned}$$

can be written in matrix form as:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} \quad \text{that is } \underline{\underline{\mathbf{A} \cdot \mathbf{x} = \mathbf{b}}}$$

$\mathbf{A} \quad \mathbf{x} = \mathbf{b}$

Programme 5: Matrices

1. Inverse matrix method

Since:

$$\mathbf{Ax} = \mathbf{b} \quad \text{then}$$

$$\underbrace{\mathbf{A}^{-1}\mathbf{A}}_{\mathbf{I}}\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} \quad \text{that is}$$

$$\underline{\mathbf{I}}\mathbf{x} = \underline{\mathbf{A}^{-1}\mathbf{b}} \quad \text{and} \quad \underline{\mathbf{I}}\mathbf{x} = \underline{\mathbf{x}}$$

The solution is then:

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

Programme 5: Matrices

1. Inverse matrix method

Example

To solve the set of equations:

$$x_1 + 2x_2 + x_3 = 4$$

$$3x_1 - 4x_2 - 2x_3 = 2$$

$$5x_1 + 3x_2 + 5x_3 = -1$$

First write the set of equations in matrix form,

$$\boxed{\begin{pmatrix} 1 & 2 & 1 \\ 3 & -4 & -2 \\ 5 & 3 & 5 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix}}$$

$A \quad x = b$

$$\text{i.e. } \mathbf{A} \cdot \mathbf{x} = \mathbf{b} \quad \therefore \mathbf{x} = \mathbf{A}^{-1} \cdot \mathbf{b}$$

1. Inverse matrix method

Example cont'd

We then find the inverse of A

$$\mathbf{A}^{-1} = -\frac{1}{35} \begin{pmatrix} -14 & -7 & 0 \\ -25 & 0 & 5 \\ 29 & 7 & -10 \end{pmatrix}$$

$$\therefore \mathbf{x} = \mathbf{A}^{-1} \cdot \mathbf{b} = -\frac{1}{35} \begin{pmatrix} -14 & -7 & 0 \\ -25 & 0 & 5 \\ 29 & 7 & -10 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix},$$

$$\mathbf{x} = -\frac{1}{35} \begin{pmatrix} -70 \\ -105 \\ 140 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix}$$

$$\text{So finally } \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix} \therefore x_1 = 2; x_2 = 3; x_3 = -4$$

1. Inverse matrix method

Another Example:

Solve the following set of equations using the inverse matrix method:

$$\begin{array}{rrcr} 2x_1 & -x_2 & +3x_3 & = 2 \\ x_1 & +3x_2 & -x_3 & = 11 \\ 2x_1 & -2x_2 & +5x_3 & = 3 \end{array}$$

Solution

1. Write in the form $AX=b$

$$\begin{pmatrix} 2 & -1 & 3 \\ 1 & 3 & -1 \\ 2 & -2 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 11 \\ 3 \end{pmatrix} \therefore X = A^{-1}b$$

2. Get A^{-1}

$$\rightarrow \det(A) = 9$$

$$\rightarrow C = \begin{pmatrix} 13 & -7 & -8 \\ -1 & 4 & 2 \\ -8 & 5 & 7 \end{pmatrix}$$

$$\rightarrow C^T = \begin{pmatrix} 13 & -1 & -8 \\ -7 & 4 & 5 \\ -8 & 2 & 7 \end{pmatrix}$$

adj A

$$\rightarrow A^{-1} = \frac{1}{9} \begin{pmatrix} 13 & -1 & -8 \\ -7 & 4 & 5 \\ -8 & 2 & 7 \end{pmatrix}$$

3. Find x : $X = A^{-1}b$

$$= \frac{1}{9} \begin{pmatrix} 13 & -1 & -8 \\ -7 & 4 & 5 \\ -8 & 2 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 11 \\ 3 \end{pmatrix}$$

$$X = \frac{1}{9} \begin{pmatrix} -9 \\ 45 \\ 27 \end{pmatrix} = \begin{pmatrix} -1 \\ 5 \\ 3 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ 5 \\ 3 \end{pmatrix}$$

$x_1 = -1; x_2 = 5; x_3 = 3$

Programme 5: Matrices

2. Gaussian elimination method

Given:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

Create the augmented matrix **B**, where:

$$\mathbf{B} = \left(\begin{array}{ccccc|c} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} & b_n \end{array} \right)$$

$[A : b]$

Programme 5: Matrices

2. Gaussian elimination method

We can solve systems of equations by performing a sequence of operations on the augmented matrix

- This technique is called **Gaussian Elimination** (or sometimes **Gauss-Jordan Elimination**)

Programme 5: Matrices

2. Gaussian elimination method

Elementary row operations

An **elementary row operation** (ERO) on a matrix is one of the following three things:

- swapping two rows of the matrix
- multiplying one of the rows by a non-zero number
- adding or subtracting a multiple of one row from another (different) row

We write $C \sim D$ to mean that the matrix D can be obtained from the C by performing a finite sequence of EROs.

Programme 5: Matrices

Example:

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To solve $x_1 + 2x_2 - 3x_3 = 3$

$$2x_1 - x_2 - x_3 = 11$$

$$3x_1 + 2x_2 + x_3 = -5$$

This can be written $\begin{pmatrix} 1 & 2 & -3 \\ 2 & -1 & -1 \\ 3 & 2 & 1 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 11 \\ -5 \end{pmatrix}$

(Handwritten: A, x, =, b)

The augmented matrix becomes $\left(\begin{array}{ccc|c} 1 & 2 & -3 & 3 \\ 2 & -1 & -1 & 11 \\ 3 & 2 & 1 & -5 \end{array} \right)$

(Handwritten: (A:b))

$$\left(\begin{array}{ccc|c} 1 & 2 & -3 & 3 \\ 2 & -1 & -1 & 11 \\ 3 & 2 & 1 & -5 \end{array} \right)$$

Programme 5: Matrices

Example:

Now subtract $\frac{2}{1}$ times the first row from the second row
and $\frac{3}{1}$ times the first row from the third row.

This gives $\left(\begin{array}{ccc|c} 1 & 2 & -3 & 3 \\ 0 & -5 & 5 & 5 \\ 0 & -4 & 10 & -14 \end{array} \right)$ $R_2 - 2R_1$
 $R_3 - 3R_1$

Now subtract $\frac{-4}{-5}$, i.e. $\frac{4}{5}$, times the second row from the third row.

The matrix becomes $\left(\begin{array}{ccc|c} 1 & 2 & -3 & 3 \\ 0 & -5 & 5 & 5 \\ 0 & 0 & 6 & -18 \end{array} \right)$ $R_3 - \frac{4}{5}R_2$

Programme 5: Matrices

Note that as a result of these steps, the matrix of coefficients of x has been reduced to an upper triangular matrix, which is what we wanted..

Finally, we detach the right-hand column back to its original position:

$$\begin{pmatrix} 1 & 2 & -3 \\ 0 & -5 & 5 \\ 0 & 0 & 6 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ -18 \end{pmatrix} \Rightarrow \begin{aligned} x_1 + 2x_2 - 3x_3 &= 3 \\ -5x_2 + 5x_3 &= 5 \\ 6x_3 &= -18 \end{aligned}$$

$\overset{A}{\quad} \cdot \overset{x}{\quad} = \overset{b}{\quad}$

Then, by 'back-substitution', starting from the bottom row we get:

$$\begin{aligned} 6x_3 &= -18 \quad \therefore x_3 = -3 \\ -5x_2 + 5x_3 &= 5 \quad \therefore -5x_2 = 5 + 15 = 20 \quad \therefore x_2 = -4 \\ x_1 + 2x_2 - 3x_3 &= 3 \quad \therefore x_1 - 8 + 9 = 3 \quad \therefore x_1 = 2 \\ \therefore \underline{x_1 = 2}; \underline{x_2 = -4}; \underline{x_3 = -3} \end{aligned}$$

3. Row echelon form and Reduced row echelon form

A matrix is said to be in **row echelon form** if:

- Any rows containing only 0s are the last rows of the matrix
- the first non-zero entry in every row is a 1 (the leading 1)
- As you move down the rows of the matrix, the leading 1s move progressively to the right.

Definition. Saying that a matrix is in **reduced row echelon form** means that it is in row echelon form and if a column contains the leading 1 of some non-zero row, then all other entries in this column are zero.

The row elimination procedure of Gaussian elimination can be continued to bring a matrix to reduced row echelon form. We notate the reduced row echelon form of a matrix A as $\mathbf{rref}(A)$.

Programme 5: Matrices

Exercises

1 Explain what is meant by row-echelon form.

2 Determine which of the following matrices are in row-echelon form:

(a) $\begin{pmatrix} 1 & 4 & 3 & 9 \\ 0 & 1 & -3 & 6 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & 2 & 5 & 6 \\ 1 & 0 & 3 & 6 \\ 0 & 0 & 1 & 4 \end{pmatrix}$

(d) $\begin{pmatrix} 1 & -1 & 3 & 7 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 5 & 6 \end{pmatrix}$

(e) $\begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & 3 \end{pmatrix}$

Answers on pg 597 of core textbook 2: Croft/ Davison

Programme 5: Matrices

Example:

Use Gaussian elimination to solve the set of equations below by reducing the matrix to row echelon form.

$$2x + y + 2z = 8$$

$$x - 3y + 3z = -4$$

$$4x + 2y - z = 1$$

Solution:

The augmented matrix is

$$\left(\begin{array}{ccc|c} 2 & 1 & 2 & 8 \\ 1 & -3 & 3 & -4 \\ 4 & 2 & -1 & 1 \end{array} \right)$$

We now apply a series of EROs to reduce the matrix to row-echelon form.

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Worked examples and exercises are in the text



Example:

Interchange the first and second row

$$\left(\begin{array}{ccc|c} 1 & -3 & 3 & -4 \\ 2 & 1 & 2 & 8 \\ 4 & 2 & -1 & 1 \end{array} \right)$$

Subtract 2x the first row from the second row, subtract 4x the first row from the third row. We get

$$\left(\begin{array}{ccc|c} 1 & -3 & 3 & -4 \\ 0 & 7 & -4 & 16 \\ 0 & 14 & -13 & 17 \end{array} \right)$$

Subtract 2x the second row from the third row to get

$$\left(\begin{array}{ccc|c} 1 & -3 & 3 & -4 \\ 0 & 7 & -4 & 16 \\ 0 & 0 & -5 & -15 \end{array} \right)$$

Example:

Divide the second row by 7, divide the third row by -5.

$$\left(\begin{array}{ccc|c} 1 & -3 & 3 & -4 \\ 0 & 1 & \frac{-4}{7} & \frac{16}{7} \\ 0 & 0 & 1 & 3 \end{array} \right)$$

The matrix is now in row-echelon form. We see that

$$z = 3$$

$$y - \frac{4}{7}z = \frac{16}{7}$$

$$x - 3y + 3z = -4$$

Using back-substitution, we get

$$z = 3$$

$$y = 4$$

$$x = -1$$

Computing inverse matrices using Gaussian elimination

By bringing an invertible matrix to reduced row echelon form, i.e. to the **identity matrix**, we can compute the matrix inverse. Given a matrix A , consider the equation

$$AA^{-1} = I$$

The method is to do row reduction on an augmented matrix which attaches the identity matrix to A . To find A^{-1} , we continue doing EROs until we obtain $\text{rref}(A) = I$.

i.e.
$$[A \mid I] \longrightarrow [I \mid X].$$

If this procedure works out, i.e. if we are able to convert A to identity using EROs, then A is invertible and $A^{-1} = X$. If we cannot obtain the identity matrix on the left, i.e. we get a row of zeros, then A^{-1} *does not exist* and A is *singular*.

We illustrate below:

$$\begin{array}{l}
 R_2 + 2R_1 \\
 R_3 + R_1
 \end{array}
 \left(\begin{array}{ccc|ccc}
 -3 & 2 & -1 & 1 & 0 & 0 \\
 6 & -6 & 7 & 0 & 1 & 0 \\
 3 & -4 & 4 & 0 & 0 & 1
 \end{array} \right) \rightarrow \left(\begin{array}{ccc|ccc}
 -3 & 2 & -1 & 1 & 0 & 0 \\
 0 & -2 & 5 & 2 & 1 & 0 \\
 0 & -2 & 3 & 1 & 0 & 1
 \end{array} \right) \rightarrow \begin{array}{l} R_3 - R_2 \\ \frac{6}{-3} = -2 \\ R_2 - (-2R_1) \\ R_2 + 2R_1 \end{array}$$

$$\left(\begin{array}{ccc|ccc}
 -3 & 2 & -1 & 1 & 0 & 0 \\
 0 & -2 & 5 & 2 & 1 & 0 \\
 0 & 0 & -2 & -1 & -1 & 1
 \end{array} \right) \xrightarrow{R_1 + R_2} \left(\begin{array}{ccc|ccc}
 -3 & 0 & 4 & 3 & 1 & 0 \\
 0 & -2 & 5 & 2 & 1 & 0 \\
 0 & 0 & -2 & -1 & -1 & 1
 \end{array} \right) \rightarrow$$

$$\begin{array}{l}
 R_1 + 2R_3 \\
 R_2 + \frac{5}{2}R_3
 \end{array}
 \left(\begin{array}{ccc|ccc}
 -3 & 0 & 0 & 1 & -1 & 2 \\
 0 & -2 & 0 & -1/2 & -3/2 & 5/2 \\
 0 & 0 & -2 & -1 & -1 & 1
 \end{array} \right) \rightarrow \left(\begin{array}{ccc|ccc}
 1 & 0 & 0 & -1/3 & 1/3 & -2/3 \\
 0 & 1 & 0 & 1/4 & 3/4 & -5/4 \\
 0 & 0 & 1 & 1/2 & 1/2 & -1/2
 \end{array} \right) \begin{array}{l} -\frac{1}{3}R_1 \\ \frac{1}{2}R_2 \\ -\frac{1}{2}R_3 \end{array}$$

I

X
X = A⁻¹

and one can check that

$$\begin{pmatrix} -3 & 2 & -1 \\ 6 & -6 & 7 \\ 3 & -4 & 4 \end{pmatrix} \begin{pmatrix} -1/3 & 1/3 & -2/3 \\ 1/4 & 3/4 & -5/4 \\ 1/2 & 1/2 & -1/2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

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Programme 5: Matrices

4. Cramer's rule

For any system $\underline{A} \underline{x} = \underline{b}$

$$x_1 = \frac{|\underline{A}_1|}{|\underline{A}|}, \quad x_2 = \frac{|\underline{A}_2|}{|\underline{A}|}, \quad \dots, \quad x_i = \frac{|\underline{A}_i|}{|\underline{A}|}$$

where $\underline{A}_i$ is obtained by replacing the i^{th} column of $\underline{A}$ with the vector $\underline{b}$.

4. Cramer's rule

For a 2×2 system,

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 &= b_1 \\ a_{21}x_1 + a_{22}x_2 &= b_2 \end{aligned}$$

Matrix Form:

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \end{Bmatrix}$$

Cramer's Rule:

$$x_1 = \frac{\begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}, \quad x_2 = \frac{\begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}$$

4. Cramer's rule: example

Example 1.1 Electrical Engineering – Currents in a circuit

The currents i_1 and i_2 in a simple circuit are connected by the equations

$$\begin{aligned} i_1 + i_2 &= 3 \\ 2i_1 + 3i_2 &= 7 \end{aligned} \quad \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$$

$A \quad X \quad = \quad b$

Calculate i_1 and i_2 using Cramer's rule.

Solution

Using Cramer's rule we have

$$\underline{i_1} = \frac{\begin{vmatrix} 3 & 1 \\ 7 & 3 \end{vmatrix}}{\begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix}} = \frac{2}{1} = \underline{\underline{2}}, \quad \underline{i_2} = \frac{\begin{vmatrix} 1 & 3 \\ 2 & 7 \end{vmatrix}}{\begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix}} = \frac{1}{1} = \underline{\underline{1}}$$

The currents in the circuit are $i_1 = 2$, $i_2 = 1$.

Example 1.2 Electrical Engineering – Circuit voltages

The voltages v_1 , v_2 and v_3 in an electric circuit are connected by the system

$$\begin{aligned}v_1 + 2v_2 + v_3 &= 2 \\2v_1 - v_2 - 2v_3 &= 5 \\2v_1 + 2v_2 + 3v_3 &= 7\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & -1 & -2 \\ 2 & 2 & 3 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 7 \end{pmatrix}$$

Use Cramer's rule to determine the three voltages.

$$v_1 = \frac{\begin{vmatrix} 2 & 2 & 1 \\ 5 & -1 & -2 \\ 7 & 2 & 3 \end{vmatrix}}{\begin{vmatrix} 1 & 2 & 1 \\ 2 & -1 & -2 \\ 2 & 2 & 3 \end{vmatrix}} = \frac{-39}{-13} = 3$$

$$v_2 = \frac{\begin{vmatrix} 1 & 2 & 1 \\ 2 & 5 & -2 \\ 2 & 7 & 3 \end{vmatrix}}{-13} = \frac{13}{-13} = -1$$

$$v_3 = \frac{\begin{vmatrix} 1 & 2 & 2 \\ 2 & -1 & 5 \\ 2 & 2 & 7 \end{vmatrix}}{-13} = \frac{-13}{-13} = 1$$

The voltages are $v_1 = 3$, $v_2 = -1$ and $v_3 = 1$.

Comparison of methods

Inverse matrix method

- This is an elementary method but it is very inefficient when the number of equations exceeds beyond three.
- Can't use if matrix is singular
- Can't use if it is not a square matrix

Gaussian elimination method

The most efficient method and should be used in most cases. It must be used when there is a singular or non-square system.

Reduced row echelon form

- Uses Gaussian elimination so same advantage as above.
- Particularly useful to compute inverses of large matrices.

Cramer's Rule

- Quickest method for 2-3 unknowns
- For four or more unknowns, the determinants become too large to make this a sensible approach.

Programme 5: Matrices

Test Exercise 5 cont'd



- 7 Express the following set of linear equations in matrix form:

$$2x_1 + 4x_2 - 5x_3 = -7$$

$$x_1 - 3x_2 + x_3 = 10$$

$$3x_1 + 5x_2 + 3x_3 = 2$$

- 8 Solve the following set of linear equations by the matrix method:

$$x_1 + 3x_2 + 2x_3 = 3$$

$$2x_1 - x_2 - 3x_3 = -8$$

$$5x_1 + 2x_2 + x_3 = 9$$



- 9 For the following set of simultaneous equations:

(a) form the augmented coefficient matrix

(b) solve the equations by Gaussian elimination.

$$x_1 + 2x_2 + 3x_3 = 5$$

$$3x_1 - x_2 + 2x_3 = 8$$

$$4x_1 - 6x_2 - 4x_3 = -2$$

Answers on page 1138 Stroud

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Programme 5: Matrices

$$y = 2x - 5$$
$$y = 2x - 3$$

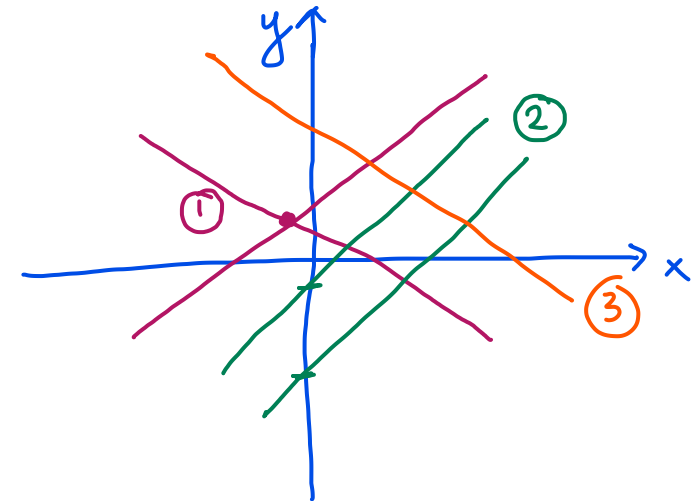
How many solutions are there?

How can we decide how many solutions there will be?

Let's take an example of a 2x2 system of equations:

$$ax + by = e$$

$$cx + dy = f$$



Assuming that at least one of a and b is not zero and at least one of c and d is not zero, each equation is that of a line in the plane. One of three things happens. ① The lines intersect at one point and the system has a unique solution. ② The lines are parallel so do not intersect and the system has no solution. ③ The two equations describe the same line and every point on that line (there are infinitely many of them) is a solution.

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Worked examples and exercises are in the text



Programme 5: Matrices

How many solutions are there?

If we have a 3×3 system of equations and at least one coefficient in each equation is non-zero, each equation is that of a plane. Again one of three things happens:

- There is a unique solution
- There is no solution
- There are infinitely many solutions

This is also true for the general $m \times n$ system

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Worked examples and exercises are in the text

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Programme 5: Matrices

General solutions:

Given the equations

$$\begin{array}{ccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \dots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \dots & + & a_{2n}x_n & = & b_2 \\ \vdots & & & & \vdots & & & & \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \dots & + & a_{mn}x_n & = & b_m \end{array}$$

the set of all possible solutions $(x_1, \dots, x_n)$ is called the **general solution**. The general solution may be empty, or may contain more than one possible set of values.

Programme 5: Matrices

An example with no solutions:

Now consider the equations with augmented matrix

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 1 \\ 4 & 2 & 3 & -1 \\ 6 & 3 & -1 & 11 \end{array} \right)$$

Performing EROs leads to the augmented matrix

$$\left(\begin{array}{ccc|c} 1 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right) \begin{array}{l} \longrightarrow x_1 + \frac{1}{2}x_2 + \overset{=0}{0}x_3 = 0 \quad \checkmark \\ \longrightarrow x_3 = 0 \quad \checkmark \\ \longrightarrow 0x_1 + 0x_2 + 0x_3 = 1 \quad \text{impossible} \\ \quad \quad \quad \underbrace{\quad\quad\quad}_{0 \neq 1} \end{array}$$

This says that $0 = 1$, which is impossible. So there are no solutions.

Programme 5: Matrices

An example with infinitely many solutions:

Consider the equations

$$v + 2w - 3x + 3y + 2z = 0$$

$$-v - w + 3x + z = 3$$

$$v + 2w + y = 1$$

$$-v - w + 4y + 5z = 4$$

$$v + 2w + x + 7y + 6z = 8$$

An example with infinitely many solutions:

Write down the augmented matrix

$$\left(\begin{array}{ccccc|c} 1 & 2 & -3 & 3 & 2 & 0 \\ -1 & -1 & 3 & 0 & 1 & 3 \\ 1 & 2 & 0 & 1 & 0 & 1 \\ -1 & -1 & 0 & 4 & 5 & 4 \\ 1 & 2 & 1 & 7 & 6 & 8 \end{array} \right)$$

Subtract multiples of row 1

$$\left(\begin{array}{ccccc|c} 1 & 2 & -3 & 3 & 2 & 0 \\ 0 & 1 & 0 & 3 & 3 & 3 \\ 0 & 0 & 3 & -2 & -2 & 1 \\ 0 & 1 & -3 & 7 & 7 & 4 \\ 0 & 0 & 4 & 4 & 4 & 8 \end{array} \right)$$

An example with infinitely many solutions:

Subtract multiples of row 2

$$\left(\begin{array}{ccccc|c} 1 & 2 & -3 & 3 & 2 & 0 \\ 0 & 1 & 0 & 3 & 3 & 3 \\ 0 & 0 & 3 & -2 & -2 & 1 \\ 0 & 0 & -3 & 4 & 4 & 1 \\ 0 & 0 & 4 & 4 & 4 & 8 \end{array} \right)$$

Divide row 5 by 4 and swap with row 3

$$\left(\begin{array}{ccccc|c} 1 & 2 & -3 & 3 & 2 & 0 \\ 0 & 1 & 0 & 3 & 3 & 3 \\ 0 & 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & -3 & 4 & 4 & 1 \\ 0 & 0 & 3 & -2 & -2 & 1 \end{array} \right)$$

An example with infinitely many solutions:

Subtract multiples of row 3

$$\left(\begin{array}{ccccc|c} 1 & 2 & -3 & 3 & 2 & 0 \\ 0 & 1 & 0 & 3 & 3 & 3 \\ 0 & 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 7 & 7 & 7 \\ 0 & 0 & 0 & -5 & -5 & -5 \end{array} \right)$$

Divide the bottom rows by 7 and -5 and subtract row 4 from row 5

$$\left(\begin{array}{ccccc|c} 1 & 2 & -3 & 3 & 2 & 0 \\ 0 & 1 & 0 & 3 & 3 & 3 \\ 0 & 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

An example with infinitely many solutions:

The final result is a matrix in Reduced Echelon Form as follows.

$$\left(\begin{array}{ccccc|c} \textcircled{1} & 0 & 0 & 0 & -1 & 0 \\ 0 & \textcircled{1} & 0 & 0 & 0 & 0 \\ 0 & 0 & \textcircled{1} & 0 & 0 & 1 \\ 0 & 0 & 0 & \textcircled{1} & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

Handwritten annotations: 'v' above the first 1, 'w' above the second 1, 'x' above the third 1, and 'y' above the fourth 1.

Converting back to equations gives us

$$\begin{aligned} v - z &= 0 \\ w &= 0 \\ x &= 1 \\ y + z &= 1 \\ 0 &= 0 \end{aligned}$$

The final equation doesn't provide any information, so we can't pin down the value of z . We'll just suppose it has some value $\alpha \in \mathbb{R}$.

An example with infinitely many solutions:

Re-arranging the equations that involve z now gives us the general solution:

$$v = \alpha$$

$$w = 0$$

$$x = 1$$

$$y = 1 - \alpha$$

$$z = \alpha$$

where $\alpha \in \mathbb{R}$. Notice that we can also use this solution to work out those solutions which only involve integers if we're asked to do so.

Another Example

[illegible]

An example with infinitely many solutions:

The corresponding equations define x_2 , x_4 , x_5 , x_6 and x_9 . The others we set to arbitrary values.

$$x_2 + 2x_3 - x_7 + x_8 = 5$$

$$x_4 + 2x_7 + 3x_8 = 4$$

$$x_5 - 2x_7 - 3x_8 = 3$$

$$x_6 - x_7 - x_8 = 2$$

$$x_9 = 1$$

$$x_9 = 1 \quad x_8 = \alpha \quad x_7 = \beta \quad x_6 = 2 + \beta + \alpha \quad x_5 = 3 + 2\beta + 3\alpha$$

$$x_4 = 4 - 2\beta - 3\alpha \quad x_3 = \gamma \quad x_2 = 5 - 2\gamma + \beta - \alpha \quad x_1 = \delta$$

An example with infinitely many solutions:

Remark. In order to have a consistent procedure for expressing the solutions in the case of infinitely many solutions, put the augmented matrix in row echelon form. The columns 1 through n where the leading 1s of the non-zero rows occur are called **pivot columns**. The corresponding unknowns are called **pivot unknowns**. The other unknowns are called **free unknowns**. Assign letters ($\alpha, \beta, \gamma, \delta, \dots$) in this order to the free unknowns and express the pivot unknowns in terms of them.

Programme 5: Matrices

Review Summary

1. *Singular* square matrix: $|A| = 0$
2. *Non-singular* square matrix $|A| \neq 0$.

3. *Elementary operations and equivalent matrices*

Each of the following row operations on matrix A produces a *row equivalent matrix* B where the order and rank of B are the same as those of A . We write $A \sim B$.

- (1) Interchanging two rows
- (2) Multiplying each element of a row by the same non-zero scalar quantity
- (3) Adding or subtracting corresponding elements from those of another row.

These operations are called *elementary row operations*.

STROUD

Worked examples and exercises are in the text

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Programme 5: Matrices

Review Summary

4. *Solution of sets of equations*

(a) *Inverse matrix method* $Ax = b; \quad x = A^{-1}b$

To find A^{-1}

(1) evaluate $|A|$

(2) form C , the matrix of cofactors of A

(3) write C^T , the transpose of A

(4) $A^{-1} = \frac{1}{|A|} \times C^T.$

(b) *Gaussian elimination method* $Ax = b$

(1) form augmented matrix $[A|b]$

(2) operate on rows to convert to $[U|b']$ where U is the upper-triangular matrix.

(3) expand from bottom row to obtain x .

Review Summary:

(c) A matrix is said to be in **row echelon form** if:

- Any rows containing only 0s are the last rows of the matrix
- the first non-zero entry in every row is a 1 (the leading 1)
- As you move down the rows of the matrix, the leading 1s move progressively to the right.

Saying that a matrix is in **reduced row echelon form** means that it is in row echelon form and if a column contains the leading 1 of some non-zero row, then all other entries in this column are zero.

5. Using Gaussian elimination to compute inverses (by reducing to row-echelon form up to the identity matrix.)

(1) form the combined coefficient matrix $[A|I]$

(2) row transformations to convert to $[I|A^{-1}]$

6. Cramer's rule:

$$x_1 = \frac{\begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}, \quad x_2 = \frac{\begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}$$